

Week 6 Weekly Report

Week 6 trained, tuned, and compared the two machine-learning approaches. Cross-validation used a 20-day gap, candidate selection used the separate validation block, and the 252-row test period was evaluated once afterward.

1 Work Completed

- Tuned linear, Random Forest, XGBoost, and neural-network candidates for volatility and direct-price tasks.
- Selected each route by validation RMSE after purge-aware cross-validation tuning.
- Evaluated BSM, ML volatility plus BSM, and direct supervised pricing using MAE, RMSE, R^2 , and mean error.
- Measured results by VIX regime to identify unstable periods.
- Generated SHAP and permutation-importance outputs for the exact selected models.
- Saved the selected model bundles and metadata for Week 7 and Week 8 reuse.

2 Deliverables

- Week6_Model_Training_Comparison.ipynb
- model_training.py
- training_config.json
- trained_models/best_volatility_model.pkl and trained_models/best_direct_pricing_model.pkl

joblib runtime copies and model_bundle_metadata.json

- MODEL_CARD.md
- metrics, predictions, SHAP tables, and figures
- Model Performance Report

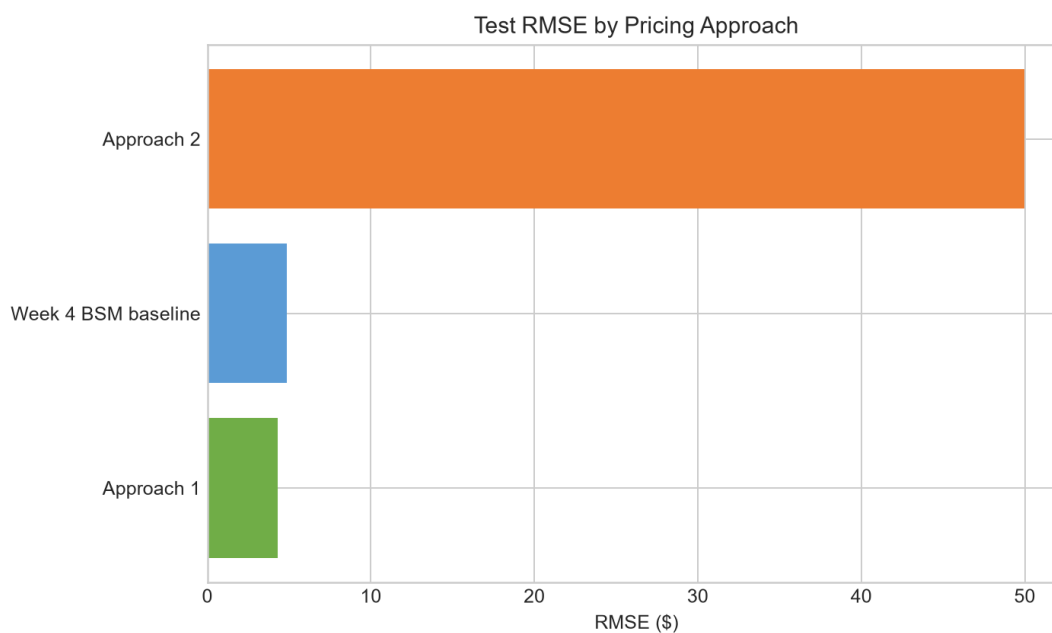
3 Progress Summary

Approach 1, a Random Forest volatility forecast passed through BSM, produced the lowest test RMSE of 4.3203. This is 11.30% below the Week 4 BSM RMSE of 4.8706. The direct Approach 2 model failed under the later price regime, with RMSE of 49.9891. All routes use the same proxy target and test dates.

4 Experimental Results

Pricing route	MAE	RMSE	R ²	Role
Approach 1 RF vol plus BSM	3.1384	4.3203	0.9575	Selected
Week 4 BSM baseline	3.5698	4.8706	0.9460	Benchmark
Approach 2 direct price	45.0275	49.9891	-4.6915	Failed OOD

Figure 1 Held-out test RMSE comparison



5 Issues Encountered

- The test target is an ex-post forward-volatility chooser proxy, so improved proxy error does not prove improved performance against actual chooser transactions.
- The direct price model extrapolates poorly beyond the development price range. Week 8 retains BSM and the selected volatility route and displays an out-of-range warning.
- SHAP values describe model behavior on the analyzed sample; they do not establish causal effects.

6 Plan for Week 7

- Run the required volatility and rate stress tests.
- Quantify the influence of VIX-related features.
- Build the public-data update path and Streamlit prototype.